

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show / Hide Password Visibility Toggle

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle in the Login Module
Student Name	AAM. Nusky
Student ID	JF/ICT/24/01
Implementation Status	Completed
Date	8 October 2026

1. Change Request Summary

1.1 Business Requirement

Users previously entered passwords into a permanently masked input field without an option to verify the entered text. The change provides an intuitive password visibility toggle so users can verify credentials before submission.

1.2 Existing Behaviour

Before implementation:

- LoginFormSection.jsx used type="password".
- Password characters were always masked.
- No visibility toggle button or eye indicator existed.
- Existing login submission passed credentials through the existing onSubmit flow.
- The Student Form password field was also statically masked.

1.3 Requested Change

- Toggle input type between "password" and "text".
- Default to masked state on component load.
- Display eye / eye-slash indicators.
- Add aria-label and aria-pressed.
- Use type="button" for the visibility control.
- Add layout spacing so password text does not overlap the icon.
- Support keyboard operation through the native button.
- Preserve existing validation, error handling, submission and authentication flows.

2. Implementation

2.1 Implementation Summary

CR-016 was implemented as a frontend enhancement. LoginFormSection.jsx now uses a showPassword state to dynamically switch the password field between masked and visible states. The same show/hide behavior was also applied to the Student Form password field. Existing login, student validation, API submission and authentication behavior were preserved.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/LoginFormSection.jsx	Modified	Added show/hide state, dynamic input type, toggle button and accessibility attributes.
frontend/src/styles/login.css	Modified	Added password wrapper, right padding, toggle positioning, focus, hover and disabled styling.
StudentFormModal.jsx	Modified	Added password visibility state and toggle while preserving validation and API logic.

Additional files: None.

2.3 Technical Changes

Password Visibility State

- Added showPassword state initialized to false.
- False keeps the password masked on initial render.
- Toggle action switches the state between true and false.

Dynamic Input Type

- showPassword=false → type="password".
- showPassword=true → type="text".
- The password value itself is not modified by the toggle.

Accessibility and Form Behaviour

- aria-label changes between Show password and Hide password.
- aria-pressed reflects the current visibility state.
- type="button" prevents accidental form submission.
- Native button behavior supports keyboard focus and Enter/Space activation.

Layout and Styling

- Password wrapper provides positioned layout.
- Input receives right padding to avoid icon overlap.
- Toggle is positioned on the right side of the field.
- Focus-visible, hover and disabled states are styled.

Existing Flow Preservation

- Login continues to call onSubmit(username.trim(), password).
- Existing loading and error handling remain unchanged.
- Student password validation remains unchanged.
- Student create/update API behavior remains unchanged.
- No database, backend authentication, token or hashing changes were made.

2.4 Implementation Logic

1. Initialize showPassword as false.
2. Render password field masked by default.
3. Provide eye toggle button beside the field.

4. Invert showPassword when the button is activated.
5. Switch input type between password and text.
6. Update icon and accessibility state.
7. Keep existing password value and submit handler unchanged.
8. Apply equivalent behavior to Student Form password field.

2.5 Error and Data Handling

- Existing Login error banners remain unchanged.
- Existing Student Form validation remains unchanged.
- Toggling visibility does not clear or modify the password value.
- No password is persisted to browser storage by this enhancement.
- Backend authentication and verification remain unchanged.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend testing.

- Functional testing
- Positive testing
- Accessibility / keyboard testing
- UI testing
- Regression testing
- Validation and error testing

Environment: Development

Browser: Chrome / Edge / Firefox

Test Role: Administrator / Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Login page	Password masked by default	Password masked	Pass
TC-02	Click eye icon	Password becomes visible	Password visible	Pass
TC-03	Click eye-slash icon	Password becomes masked	Password masked	Pass
TC-04	Use Enter / Space on toggle	Visibility toggles without submission	Toggle operates	Pass
TC-05	Verify aria-label / aria-pressed	Accessibility state is correct	Implemented	Pass
TC-06	Submit valid Login credentials	Existing login works	Existing flow preserved	Pass
TC-07	Trigger Login error	Error banner remains	Preserved	Pass
TC-08	Open Student Form	Password masked by default	Password masked	Pass

TC-09	Toggle Student Form password	Hidden/visible switching works	Toggle operates	Pass
TC-10	Submit Student Form	Existing validation/API works	Existing flow preserved	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	10
Passed	10
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

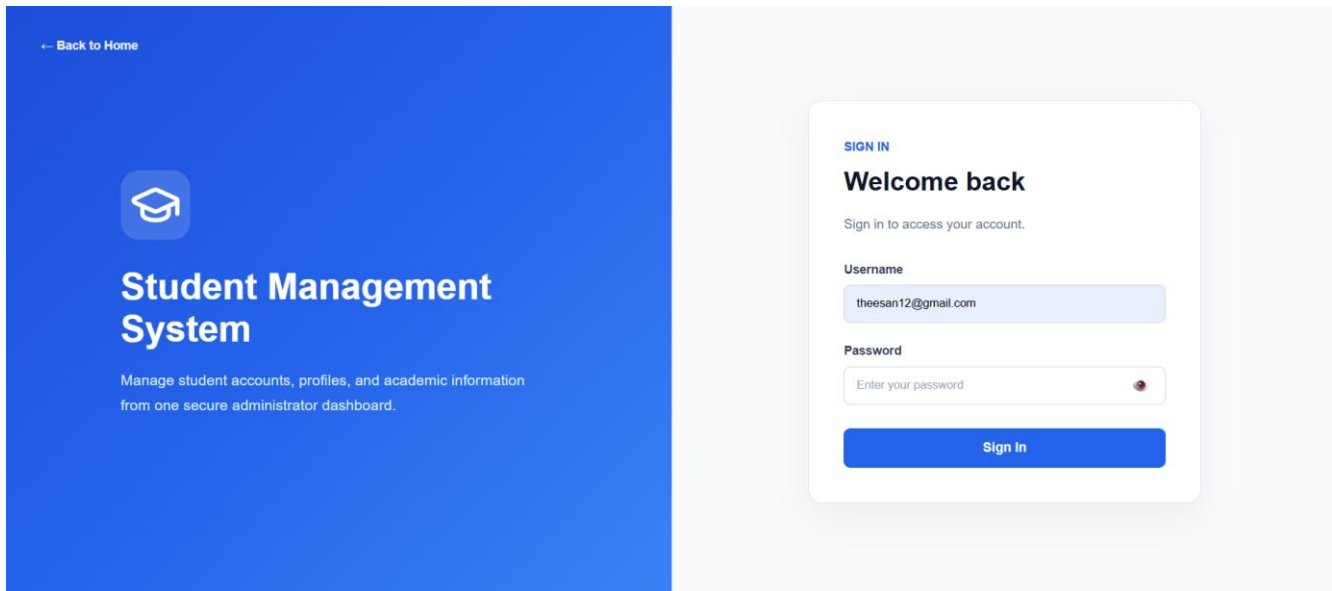
ID	Acceptance Criterion	Verification	Status
AC-001	Users can toggle password visibility	Functional toggle test	Met
AC-002	Password is masked on load	Initial-load test	Met
AC-003	Icon reflects current state	Eye / eye-slash verification	Met
AC-004	Toggle does not submit form	type="button" verification	Met
AC-005	Keyboard accessible and has aria-label	Keyboard/accessibility verification	Met
AC-006	Password text is not obscured by icon	UI/layout verification	Met
AC-007	Valid credentials work in either state	Login regression test	Met
AC-008	Existing validation and role flows continue	Regression verification	Met

5. Evidence

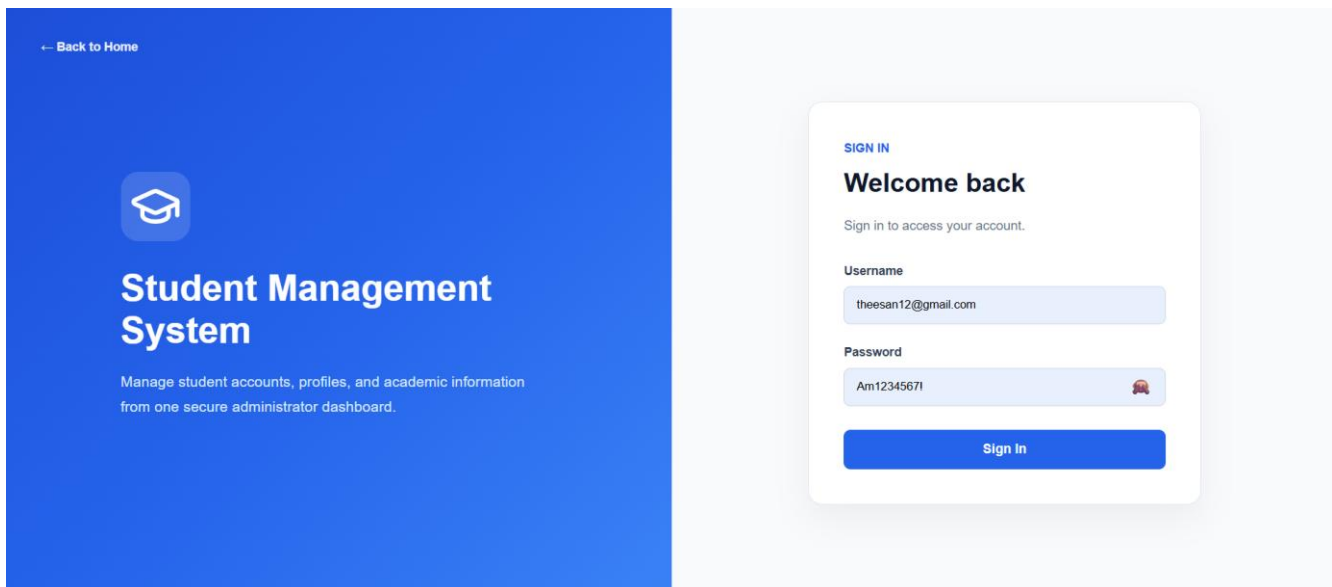
A maximum of two relevant screenshots should be attached.

Evidence 01 – Login Password Visibility

Related: AC-001, AC-002, AC-003, AC-006



Evidence 02 – Student Form Password Visibility
Related: AC-001, AC-002, AC-006



6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing Login Flow	Verified / Preserved
Existing Student Validation & Submission	Verified / Preserved
Final Status	Accepted

6.1 Final Summary

CR-016 was implemented as a frontend password visibility enhancement. The Login Form now allows users to switch between masked and visible password states with a masked default, dynamic eye indicators, accessibility

attributes, keyboard accessibility and protected input spacing. The same show/hide behavior was applied to the Student Form password field. Existing validation, login submission, authentication, student submission and error handling were preserved. No backend or database changes were required.

6.2 Issues / Limitations

No significant issues identified for the implemented password visibility functionality.

Student Name: AAM. Nusky

Student ID: JF/ICT/24/01

Date: 8 October 2026

Signature: _____

~ END OF REPORT ~